DERIVED COMPLETION HAS FINITE COHOMOLOGICAL DIMENSION BOUNDED BY THE NUMBER OF GENERATORS OF THE IDEAL

THE STACKS PROJECT AUTHORS

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Editorial note. The printed text of part (2) concludes that the cohomology of the completion of the cone vanishes below degree r while the statement predicts the shift minus r; the printed text is copied exactly as it stands and no correction is applied, since the lane delivers the author's own words.

Editorial note. The proof refers twice to earlier results of this chapter, namely the Koszul description of the completion and the exact sequence for the cohomology of a derived limit; both are navigation stubs with their roles recorded in the reasons.

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Allowed standard prerequisites (editorial).

- Derived completion of an object is the derived limit of its derived tensor products with the Koszul complexes on the powers of a generating set. invoked at proof lines 26525-26528. Turns the completion into a derived limit so that its cohomology can be computed from the Koszul system.
- For a system of complexes there is a short exact sequence expressing the cohomology of the derived limit through the first derived limit and the limit of the cohomologies. invoked at proof lines 26528-26532. Provides the displayed short exact sequence from which the vanishing of the cohomology of the completion is read off.
- The Koszul complex on the n-th powers of r generators sits in degrees from -r to 0, so tensor products with it vanish in the corresponding range of degrees. used at proof lines 26537-26541 and 26547-26550. Gives the degree ranges in which the cone of the given morphism has vanishing cohomology after tensoring with the Koszul system, which pins down the vanishing range for the completion.

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Source reference. Original lemma 0920: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26388–26388. The reference gives the description of the completion as a derived limit of Koszul complexes, used at proof lines 26525-26527; the exact sequence and the two case analyses that follow are inside the counted span.

Source reference. Original lemma 0CQE: sources/stacks-more-algebra/pinned-original-source.tex, source lines 23653–23653. The reference supplies the displayed short exact sequence for the cohomology of the completion, invoked at proof lines 26528-26532; everything read off from that sequence is inside the counted span.

Lemma 0.2. *Let A be a ring and let $I \subset A$ be an ideal which can be generated by r elements. Then derived completion has finite cohomological dimension:*

1. *Let $K \to L$ be a morphism in $D(A)$ such that $H^i(K) \to H^i(L)$ is an isomorphism for $i \geq 1$ and surjective for $i = 0$. Then $H^i(K^\wedge) \to H^i(L^\wedge)$ is an isomorphism for $i \geq 1$ and surjective for $i = 0$.*
2. *Let $K \to L$ be a morphism of $D(A)$ such that $H^i(K) \to H^i(L)$ is an isomorphism for $i \leq -1$ and injective for $i = 0$. Then $H^i(K^\wedge) \to H^i(L^\wedge)$ is an isomorphism for $i \leq -r - 1$ and injective for $i = -r$.*

Proof. Say I is generated by $f_1, \ldots, f_r$. For any $K \in D(A)$ by Lemma 0920 we have $K^\wedge = R\lim K \otimes^{\mathbf{L}}_A K_n$ where K_n is the Koszul complex on $f_1^n, \ldots, f_r^n$ and hence we obtain a short exact sequence

$$0 \to R^1 \lim H^{i-1}(K \otimes^{\mathbf{L}}_A K_n) \to H^i(K^\wedge) \to \lim H^i(K \otimes^{\mathbf{L}}_A K_n) \to 0$$

by Lemma 0CQE.

Proof of (1). Pick a distinguished triangle $K \to L \to C \to K[1]$. Then $H^i(C) = 0$ for $i \geq 0$. Since K_n is sitting in degrees ≤ 0 we see that $H^i(C \otimes^{\mathbf{L}}_A K_n) = 0$ for $i \geq 0$ and that $H^{-1}(C \otimes^{\mathbf{L}}_A K_n) = H^{-1}(C) \otimes_A A/(f_1^n, \ldots, f_r^n)$ is a system with surjective transition maps. The displayed equation above shows that $H^i(C^\wedge) = 0$ for $i \geq 0$. Applying the distinguished triangle $K^\wedge \to L^\wedge \to C^\wedge \to K^\wedge[1]$ we get (1).

Proof of (2). Pick a distinguished triangle $K \to L \to C \to K[1]$. Then $H^i(C) = 0$ for $i < 0$. Since K_n is sitting in degrees $-r, \ldots, 0$ we see that $H^i(C \otimes^{\mathbf{L}}_A K_n) = 0$ for $i < -r$. The displayed equation above shows that $H^i(C^\wedge) = 0$ for $i < r$. Applying the distinguished triangle $K^\wedge \to L^\wedge \to C^\wedge \to K^\wedge[1]$ we get (2). □